

01 THE PROBLEM

Small models fail on procedure

7B and 8B open-weight models, investigating alone

0.083

Qwen2.5-7B: share verified

probes until the budget runs out; a verdict in 6 of 72 cases



0 / 3,389

Llama-3.1-8B: verdicts / decisions

always asks for one more probe, even at $p > 0.999$



0.069

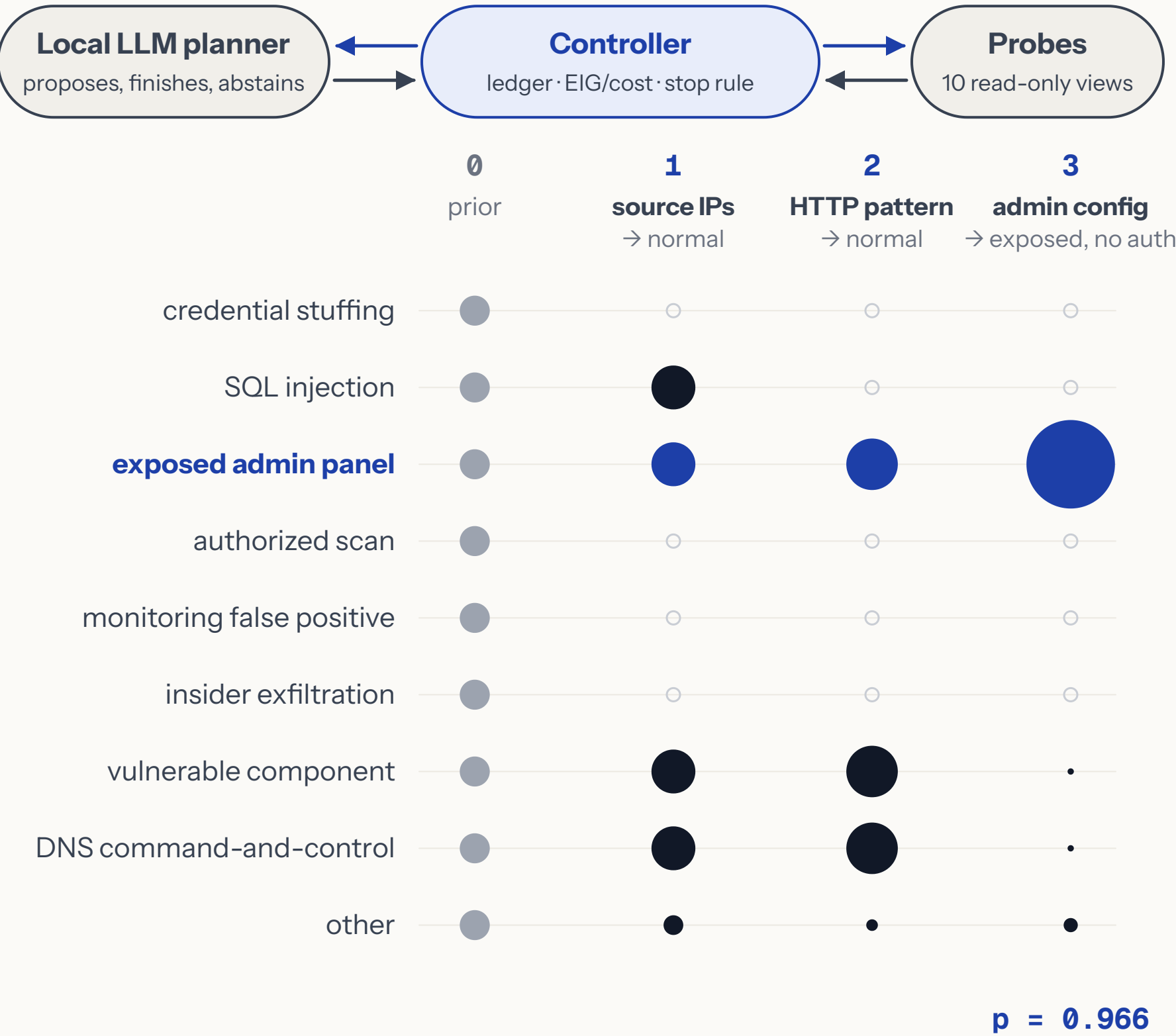
Qwen2.5-7B with its own tables

cannot write calibrated probe-outcome likelihoods



02 THE CONTROLLER

Hypotheses eliminated, probe by probe



The controller stops and cites evidence; the verifier confirms “exposed admin panel”.
Real episode sec-base-02, stopping study. Circle area = posterior; rings = eliminated.

03 WHAT WE MEASURED

Four pre-registered studies

5 models, 2 families, 7,272 audited episodes

- **No oracle needed**
0.125 → 1.000
Tables counted from LLM-free runs work as well as the true generator (Qwen 7B)
- **The ranking helps every model**
+0.26 to +0.35
EIG/cost vs random, same planner prompt; taking choice away: 7B +0.56, 72B -0.28
- **What to probe ≠ when to stop**
0 → 0.917
A controller stop rescues Llama 8B; probe choice adds nothing detectable after
- **The controller alone comes close**
0.917 vs 1.000
No LLM vs the best LLM configuration: strong models wait for confirming evidence
- **Public SOC data cannot test this**
3 of 3
ExCyTIn, GUIDE and OTRF: the alert names the answer, or labels follow policy

Put the procedure outside the model.

What remains for the model: judging when evidence suffices, and reading raw telemetry into the observations the controller consumes.